Task Initialization

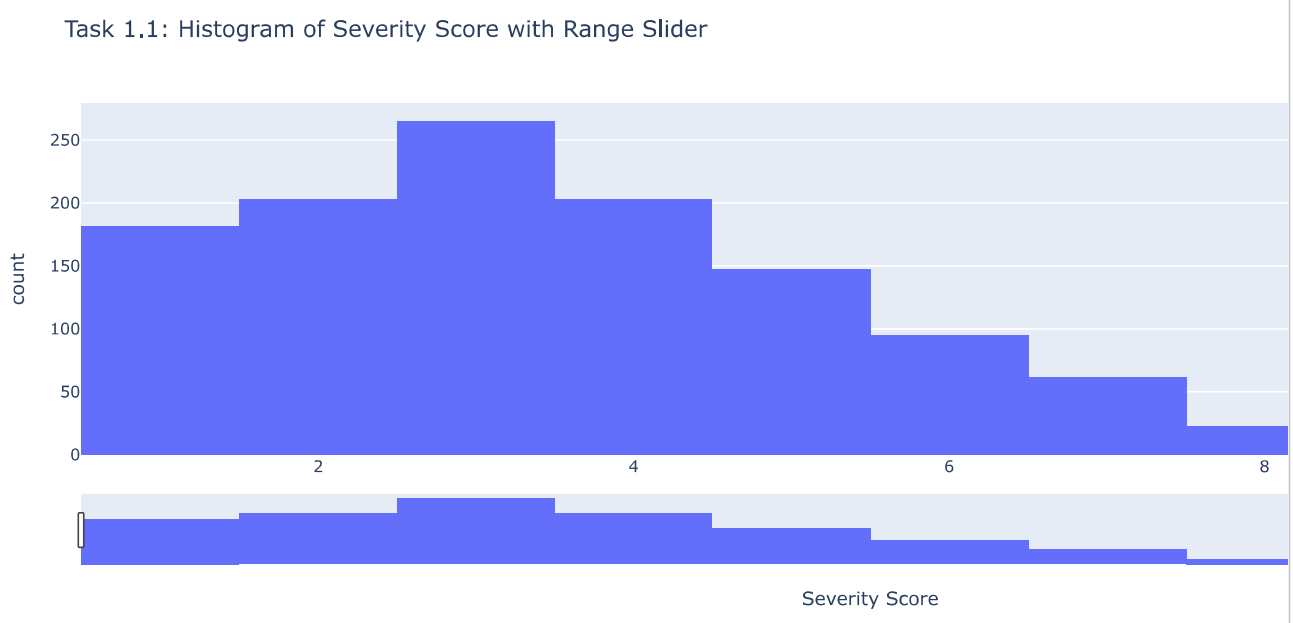
Let's first inspect the screenshot to understand the tasks, and display the first few rows of the CSV dataset.

Task 1.1: Histogram of Severity Score with Range Slider

We construct an interactive histogram of the `severity_score` column containing a range slider on the x-axis.

```
import plotly.express as px

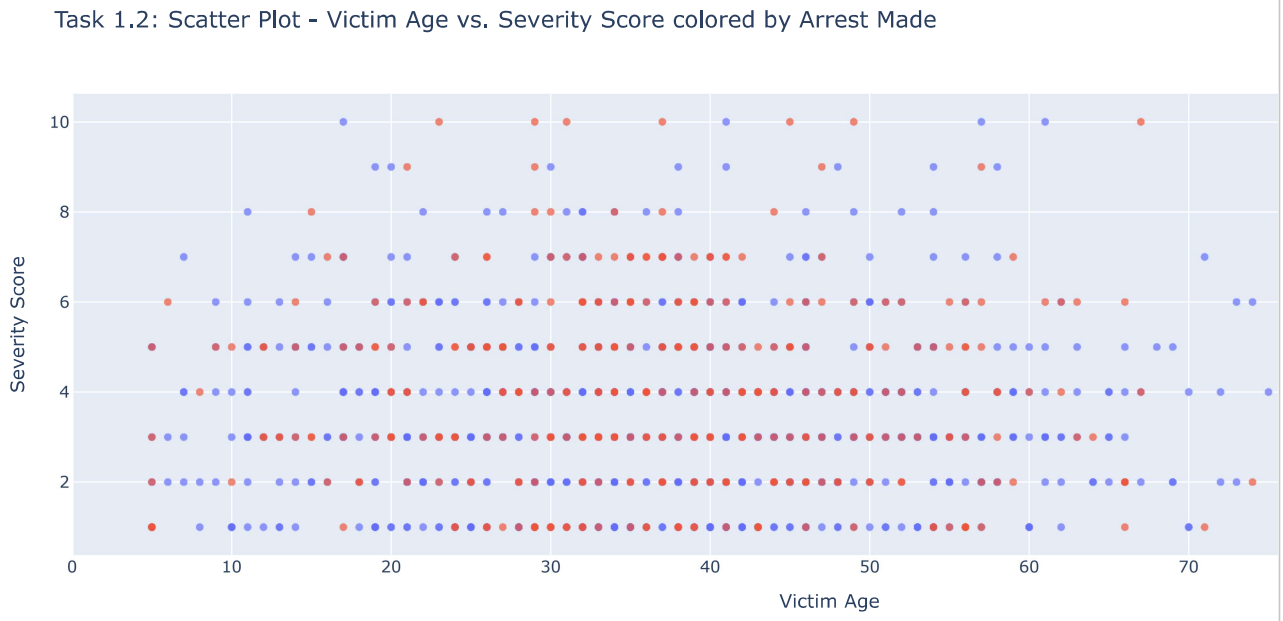
fig1 = px.histogram(
    df,
    x="severity_score",
    title="Task 1.1: Histogram of Severity Score with Range Slider",
    labels={"severity_score": "Severity Score"}
)
fig1.update_layout(xaxis_rangeslider_visible=True)
fig1.show()
```



Task 1.2: Scatter Plot (Severity Score vs. Victim Age, colored by Arrest Made)

We map this to visualising how incident severity correlates with victim age, segmenting by whether an arrest was made.

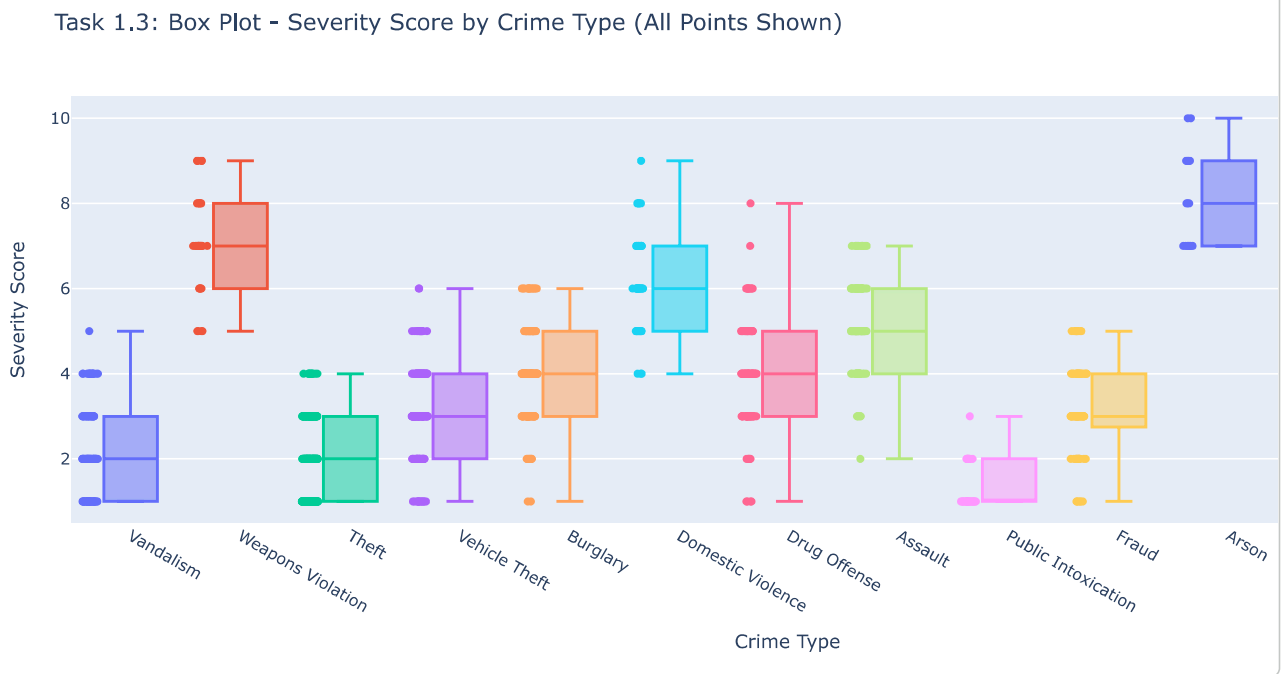
```
fig2 = px.scatter(
    df,
    x="victim_age",
    y="severity_score",
    color="arrest_made",
    title="Task 1.2: Scatter Plot - Victim Age vs. Severity Score colored by Arrest Made",
    labels={"victim_age": "Victim Age", "severity_score": "Severity Score", "arrest_made": "Arrest Made"},
    opacity=0.7
)
fig2.show()
```



Task 1.3: Box Plot (Severity Score by Crime Type with All Points Shown)

We show the distribution of severity scores for each category of crime, ensuring individual data points are fully visible.

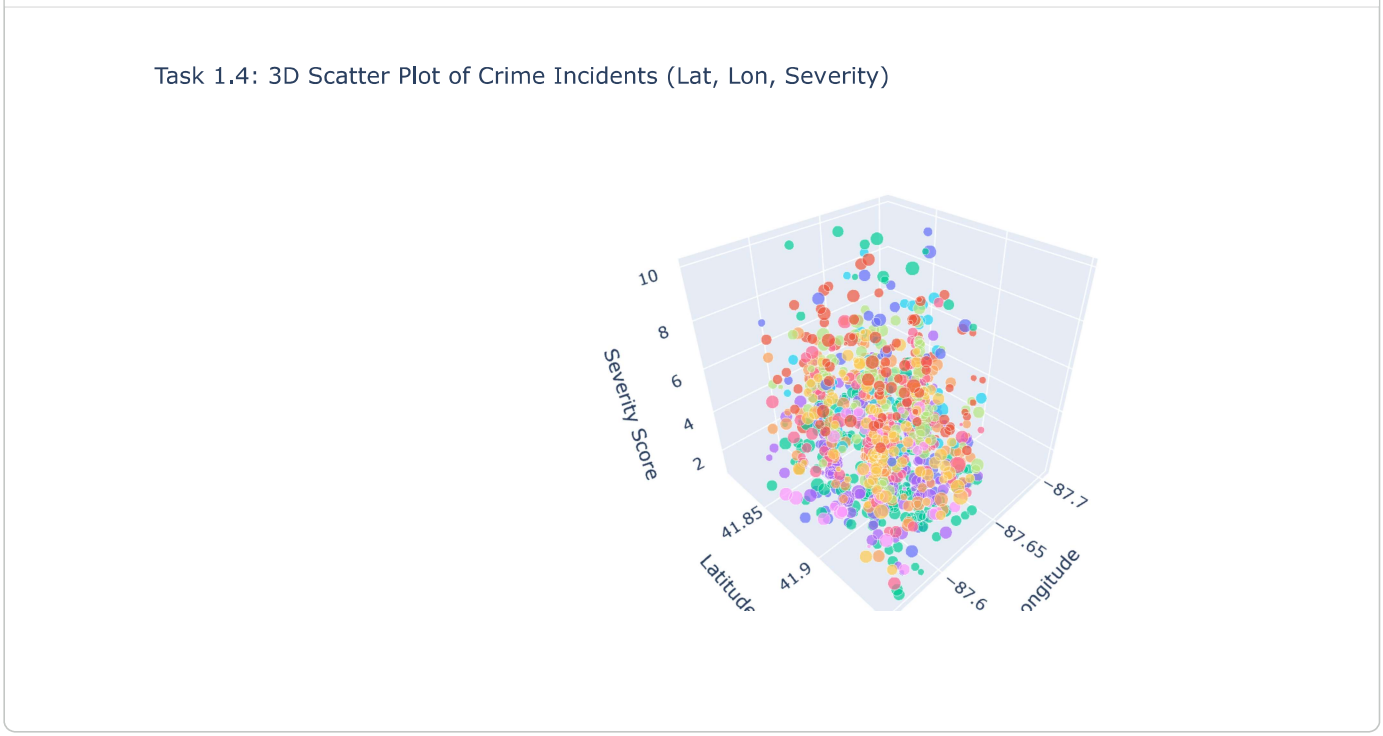
```
fig3 = px.box(
    df,
    x="crime_type",
    y="severity_score",
    color="crime_type",
    points="all",
    title="Task 1.3: Box Plot - Severity Score by Crime Type (All Points Shown)",
    labels={"crime_type": "Crime Type", "severity_score": "Severity Score"}
)
fig3.show()
```



Task 1.4: 3D Scatter Plot (Latitude, Longitude, and Severity Score)

We visualize geographic locations in 3D alongside the severity of the incidents.

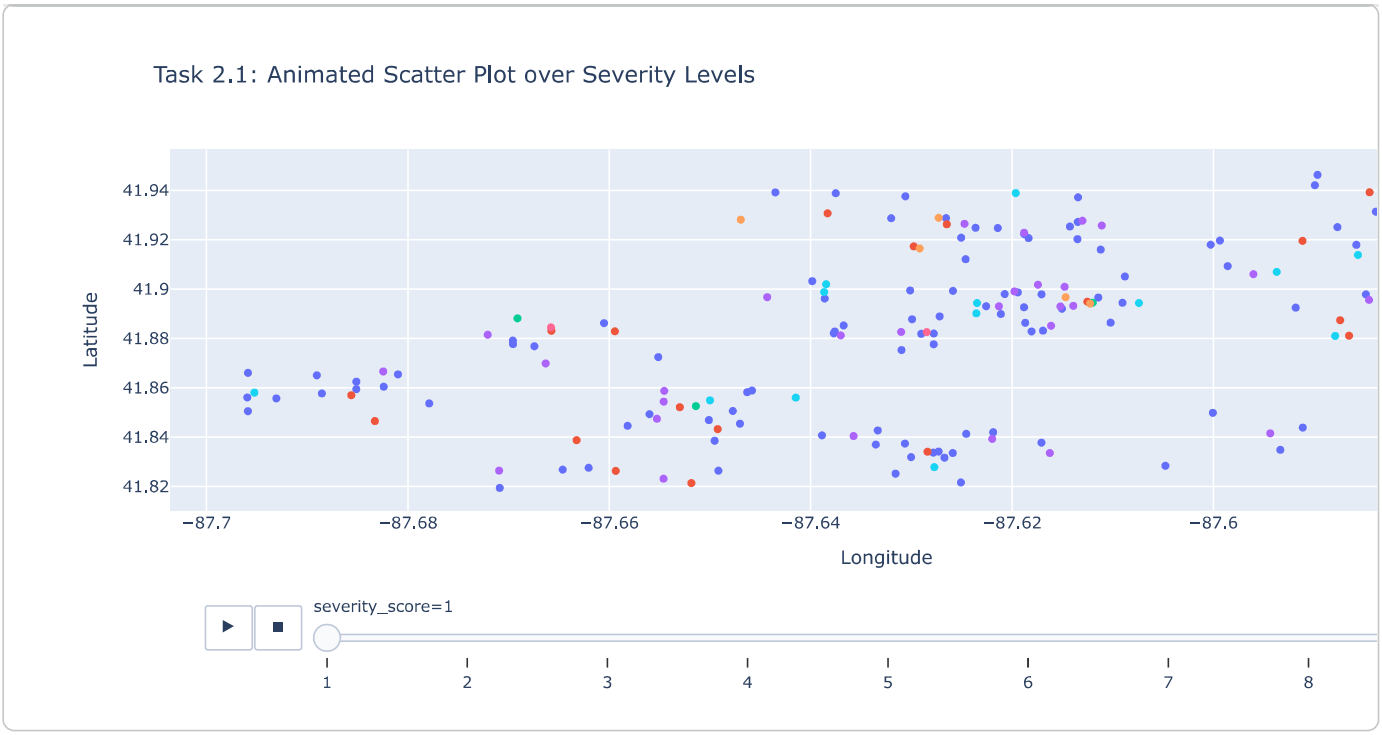
```
fig4 = px.scatter_3d(
    df.dropna(subset=["latitude", "longitude", "severity_score", "victim_age"]),
    x="longitude",
    y="latitude",
    z="severity_score",
    color="crime_type",
    size="victim_age",
    title="Task 1.4: 3D Scatter Plot of Crime Incidents (Lat, Lon, Severity)",
    labels={"longitude": "Longitude", "latitude": "Latitude", "severity_score": "Severity Score"}
)
fig4.show()
```



Task 2.1: Animated Scatter Plot (Latitude vs. Longitude animated by Severity Score)

Using the `severity_score` column as our sequence frames (sorted) to animate geographical coordinates of incidents.

```
df_sorted = df.dropna(subset=["latitude", "longitude", "severity_score"]).sort_values("severity_score")
fig5 = px.scatter(
    df_sorted,
    x="longitude",
    y="latitude",
    animation_frame="severity_score",
    animation_group="incident_id",
    color="crime_type",
    hover_name="neighborhood",
    title="Task 2.1: Animated Scatter Plot over Severity Levels",
    labels={"longitude": "Longitude", "latitude": "Latitude"}
)
fig5.show()
```



Task 2.2: Scatter Plot with Dropdown to Switch the Y-Axis Metric

We build an interactive Plotly graph where users can dynamically toggle the y-axis between `severity_score` and `victim_age`.

Gemini

```
import plotly.graph_objects as go

# Create a starting figure
fig6 = go.Figure()

# Add traces for both metrics (relative to a constant x index or latitude)
fig6.add_trace(go.Scatter(
    x=df["latitude"],
    y=df["severity_score"],
    mode='markers',
    marker=dict(color='blue'),
    name='Severity Score'
))

fig6.add_trace(go.Scatter(
    x=df["latitude"],
    y=df["victim_age"],
    mode='markers',
    marker=dict(color='green'),
    name='Victim Age',
    visible=False
))

# Create dropdown menu options
fig6.update_layout(
    updatemenus=[
        dict(
            active=0,
            buttons=list([
                dict(label="Severity Score",
                     method="update",
                     args=[{"visible": [True, False]},
                           {"yaxis": {"title": "Severity Score"}}]),
                dict(label="Victim Age",
                     method="update",
                     args=[{"visible": [False, True]},
                           {"yaxis": {"title": "Victim Age"}}])
            ]),
            direction="down",
            pad={"r": 10, "t": 10},
            showactive=True,
```

```
            x=0.1,
            xanchor="left",
            y=1.15,
            yanchor="top"
        )
    ],
    title="Task 2.2: Scatter Plot with Dropdown to Switch Y-Axis Metric",
    xaxis_title="Latitude",
    yaxis_title="Severity Score"
)
fig6.show()
```

